

Willow Herring - Curriculum Vitae

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GitHub: <https://github.com/ReFil>

Technical Skills and Experience:

- *Extensive knowledge of electronic design and manufacturing, PCB design, PCB assembly, circuit debugging, design for manufacture, supply chain management*
- *Extensive experience with KiCAD, Autodesk Fusion 360 and Autodesk EAGLE*
- *Knowledge of C/C++ for both hosted and embedded systems*
- *Experience with Zephyr RTOS, Bluetooth LE and ultra low power embedded electronics hardware design and software*
- *Experience with numerous 8 and 32 bit embedded platforms including Nordic nRF52, STM32, RP2040 and Atmel ATmega, both hardware design for these platforms and software*
- *Experience with long term software maintenance for clients*
- *Experience with analog circuit design*
- *Experience with FPGAs, system verilog and digital synthesis and design, Tanner tools, Calibre tools, Silicon layout in L-Edit*
- *Experience in designing and installing automotive electronic products, including CAN bus, CAN bus reverse engineering and ADAS systems*
- *Experience with USB, USB HID, HID over GATT, I2C, SPI, QSPI, UART protocols*
- *Experience in custom hardware interface drivers in embedded applications including multi touch trackpads and angle sensors*
- *Knowledge of Linux, Bash and Git*
- *Design, building and maintenance of 3D printers*
- *RC aircraft, control systems and software. I have also been involved in developing racing drone products for retail*

Education:

Studied Maths (A*), Further Maths (A), Physics (A) and Chemistry A level (B)

First in Electrical and Electronic engineering MEng at University of

Southampton

Currently undertaking an Electrical and Electronic engineering PhD at University of Southampton

Interests and Hobbies and extra-curricular:

Contributor to numerous open source software projects including but not limited to: “Retropilot” vehicle autonomy project, Adafruit tinyuf2 bootloader, Extensive contribution to ZMK

Have developed hardware and software to give my own car SAE level 2 autonomy and released as open source (some of the work has since been commercialised by other parties)

Reverse engineering hardware designs

Undertaken several digital logic hobby projects including a stopwatch IC

Automotive and marine electrical projects

Work Experience:

Company director of Polarity Works November 2020 to date

Since Summer 2020 I have part of a two-person team developing new and innovative computer peripherals. I have learned a lot about supply chain management, dealing with retailers in Europe, the UK and the USA, as well as manufacturing partners in China which requires a lot of logistical management

All engineering is done in house, giving me substantial experience in circuit design, PCB layout, debugging, embedded firmware development and IoT development as well as the physical engineering aspects of design for manufacture and CNC manufacturing.

We also undertake custom designs for clients where we provide a complete hardware and software package and provide end user support and software maintenance for the lifecycle of the product, giving me extensive experience in end user communication, planning non disruptive upgrade pathways and software testing.

Research Assistant for University of Southampton October 2020 to date

I have assisted in numerous different research projects at the University of Southampton, primarily an IoT project with a commercial client as well as leading the electrical installation of a PV test site at Chilbolton observatory

The IoT project entails a significant amount of experimentation to evaluate the feasibility of different power sources for an IoT device, AirFuel based wireless charging as well as PV power was considered and evaluated, included in this was the development of our own AirFuel receiver.

The laboratory based at Chilbolton is still in progress, it consists of numerous different types of PV panels on different mountings, a suite of sensors including pyranometers and reference cells, and a custom datalogging setup I designed and build that tracks and logs the maximum power available from each individual PV panel in real time. Utility constraints at the site prevented us from using a conventional grid connection so the energy harvested from the panels is used to power the datalogging system, the sensors deployed at the site and excess energy is burned off in a heater.

PGR Laboratory demonstrator October 2022 to date

I perform demonstrating work in the practical undergraduate laboratory exercises at the university of Southampton. These lab exercises cover a wide range of topics including analog and digital circuits (both in simulations and in practical experimentation), FPGAs and ASIC design and integration, C and C++ programming and communications. This has expanded my proficiency in a wide range of topics and also given me experience teaching and managing a team of other demonstrators